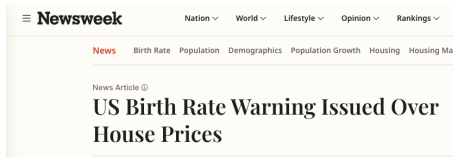


Fertility and Housing Tenure Choice

Tommaso De Santo

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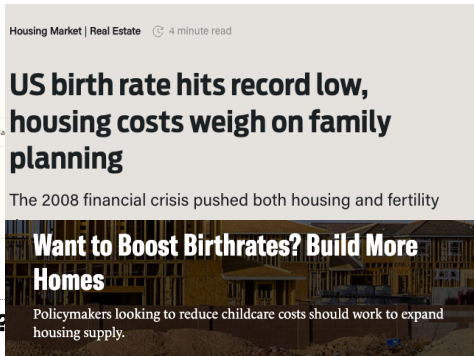
Housing and Fertility



POLICY

Will cheap housing lead to more babies?

The connection between housing and fertility rates has a missing piece.



Renting a home 'leads to lower birth rates'

Housing, Fertility, and Tenure Choice

Housing costs affect fertility, but tenure and wealth matter:

- Earlier access to homeownership increases eventual family size (Dettling & Kearney 2025, Fazio et al. 2025)
- Easier mortgage credit raises buying and births jointly; access to larger homes matters (Hacamo 2021, Couillard 2025)

Children increase how big of a house you need. This has implications for:

- Owning: tenure segmentation
- Wealth: borrowing constraints

Physical space is key. How much does it affect fertility?

What kind of housing policy is good for natality?

→ focus on space constraints and tenure choice

This Paper

Model: overlapping generations with fertility, tenure, and saving

- Childbirth raises housing needs and changes future housing demand.
- Idiosyncratic earnings risk and bequests.
- Rental size limits, down payments, and housing transaction costs.

Empirics: housing around births, birth timing, and wealth by age

Quantification: calibrate the model and study the demographic transition

- Discipline the initial economy and impose exogenous decline in fertility preferences.
- Treat today as a point in the transition to a new SS with lower population
- Compare housing policies along the path.

Model

Environment

- Discrete time, overlapping generations of households
- Adult age $a \in \{0, \dots, J\}$; survival probability s_a
 - Working age $a < a_R$; fertile window $a \in [a_F^0, a_F^1]$
 - Children do not work; they depend on their parents
- Households differ in liquid wealth, owned housing, permanent income, earnings shocks, and family composition
- Households choose:
 - **Fertility**: whether to attempt the next birth, subject to EV shocks
 - **Tenure**: rent or buy a discrete house size H_k on a housing ladder
 - **Saving**: next-period liquid wealth b_{t+1} with age-dependent borrowing limits
- Children mature stochastically; housing need depends on children at home

Preferences

n_t : children ever born; m_t : children currently at home.

Flow utility. CRRA over a consumption aggregator $\mathcal{C}$ and children at home:

$$u_t(c_t, s_t; m_t) = \frac{\mathcal{C}(c_t, s_t; m_t)^{1-\sigma}}{1-\sigma} + \psi_t m_t, \quad \sigma > 0.$$

- $\mathcal{C}$ increasing and concave in consumption and housing: $\mathcal{C}_c, \mathcal{C}_s > 0$, $\mathcal{C}_{cc}, \mathcal{C}_{ss} < 0$.
- Children raise consumption and housing needs: $\mathcal{C}_m < 0$.
- Children raise the marginal value of housing: $\partial^2 u_t / \partial s_t \partial m_t > 0$.
- $\psi_t m_t$: direct benefit of children at home.

Housing services. Households either rent or own; χ scales owner services:

$$s_t = \underbrace{\mathbf{1}\{\text{own}\} \chi h_{t+1}}_{\text{owner services}} + \underbrace{\mathbf{1}\{\text{rent}\} h_t^R}_{\text{renter services}}, \quad \chi > 0.$$

- Child costs disappear after children mature.

Bequests

Terminal utility at death:

$$B(w_{t+1}^e) = \theta_0 \frac{(\theta_1 + w_{t+1}^e)^{1-\sigma}}{1-\sigma}.$$

- θ_0 governs overall bequest motive (saving level).
- θ_1 governs its sensitivity to wealth.
- Total bequeathable wealth w_{t+1}^e includes liquid wealth and the market value of housing.

Housing, Tenure, and Budget Constraints

Income and saving.

- After-tax earnings $y_t^d(a, z_t)$ and a lump-sum transfer T_t finance consumption c_t .
- One-period bond b_{t+1} at gross rate R_b .
- No unsecured borrowing: renters cannot borrow; owners borrow only against the house.

Owning. Buy $h_{t+1} \in [0, \bar{h}]$ at price P_t .

- The house depreciates at rate δ_H and is taxed at rate τ_t^P .
- Selling costs a fraction ψ^s of the sale value.
- Buying requires a down payment; the rest is a mortgage secured by the house:

$$b_t + (1 - \psi^s)P_t h_t \geq (1 - \phi)P_t h_{t+1}, \quad b_{t+1} \geq -\phi P_t h_{t+1}.$$

Renting. Rent $h_t^R \in [0, \bar{h}^R]$ at rent r_t per unit, with $\bar{h}^R < \bar{h}$.

- The largest homes are reachable only through ownership.

Fertility and Child Aging

- During fertile ages, households choose each period whether to attempt a birth.
- The choice is discrete, with an extreme-value taste shock ε_t^F of scale κ_1 for the first birth and κ_C for later births.
- An attempt succeeds with age-specific probability π_a and adds one child. The first birth pays a one-time utility cost ξ .
- Children mature stochastically: each dependent child leaves the household with a fixed probability each period, independently of the others.

Firms and Earnings

- Competitive firms produce with effective labor:

$$Y_t = AL_t, \quad w = A.$$

- Age- and productivity-dependent earnings:

$$y_t^d(a, z_t) = \begin{cases} (1 - \tau^{SS}) w e_a z_t, & a < a_R, \\ \varpi_t, & a \geq a_R. \end{cases}$$

- e_a is effective labor supplied over the model period at age a ; z_t combines a permanent income group and a persistent earnings shock.
- Retirees receive a pay-as-you-go pension ϖ_t financed by payroll taxes; benefits adjust each period to balance the system.

Housing Supply and Price Mapping

Housing supply:

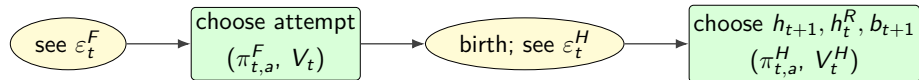
$$H_t^s = H_0 \left(\frac{P_t}{\bar{P}} \right)^\eta.$$

- H_0 : supply at reference price $\bar{P}$; η : supply elasticity.
- House prices map into user costs through

$$r_t = (R_b + \delta_H + \tau_t^P)P_t - P_{t+1}.$$

- P_t is the purchase price per room; r_t is rent per room.

The Household's Problem



State.

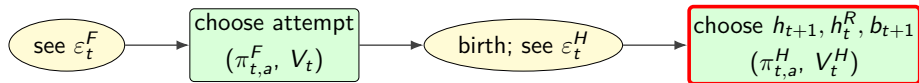
$$(b_t, h_t, z_t, a, n_t, m_t), \quad h_t \in \{0, H_1, \dots, H_K\}.$$

- b_t : liquid wealth; h_t : owner-occupied size ($h_t = 0$ if renter); z_t : earnings; a : age.
- n_t : fertility; m_t : children at home.
- A birth this period adds one to both before the housing choice.

Within the period: choose whether to attempt a birth, then $(h_{t+1}, h_t^R, b_{t+1})$.

A renter sets $h_{t+1} = 0$; an owner sets $h_t^R = 0$.

The Household's Problem: Housing and Saving



Conditional on the realized family size:

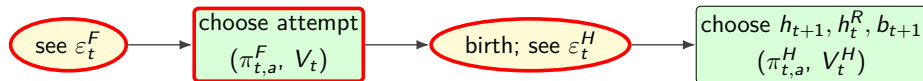
$$V_t^H(b_t, h_t, z_t, a, n, m) = \mathbb{E}_{\varepsilon_t^H} \max_{h_{t+1}, h_t^R, b_{t+1}} \left\{ u_t(c_t, \chi h_{t+1} + h_t^R; m) + \varepsilon_t^H(h_{t+1}) \right. \\ \left. + \beta \mathbb{E}_t[s_a V_{t+1}(b_{t+1}, h_{t+1}, z_{t+1}, a+1, n_{t+1}, m_{t+1}) + (1-s_a)B(w_{t+1}^e)] \right\}.$$

- Owner stock $h_{t+1} \in \{0, H_1, \dots, H_K\}$; rental flow $h_t^R \in [0, \bar{h}^R]$.

Budget and collateral.

$$c_t + b_{t+1} + r_t h_t^R + (\delta_H + \tau_t^p) P_t h_{t+1} = R_b b_t + y_t^d(a, z_t) + T_t - R_b \mathbf{1}\{h_{t+1} \neq h_t\} P_t [h_{t+1} - (1 - \psi^s) h_t], \\ b_{t+1} \geq \underline{b}_{t+1}^R \quad (h_{t+1} = 0), \quad b_{t+1} \geq \underline{b}_{t+1}^O \quad (h_{t+1} > 0).$$

The Household's Problem: Fertility Choice



At $(b_t, h_t, z_t, a, n_t, m_t)$, waiting and attempting have values

$$W_t^0 = V_t^H(b_t, h_t, z_t, a, n_t, m_t),$$

$$W_t^A = \pi_a [V_t^H(b_t, h_t, z_t, a, n_t + 1, m_t + 1) - \xi \mathbf{1}\{n_t = 0\}] \\ + (1 - \pi_a) V_t^H(b_t, h_t, z_t, a, n_t, m_t).$$

Before fertility tastes, household value is

$$V_t(b_t, h_t, z_t, a, n_t, m_t) = \mathbb{E}_{\varepsilon_t^F} \max\{W_t^0 + \varepsilon_{t,0}^F, W_t^A + \varepsilon_{t,A}^F\}.$$

Choice probabilities:

$$\pi_{t,a}^F(b_t, h_t, z_t, n_t, m_t) = \frac{\exp\{W_t^A/\kappa(n_t)\}}{\exp\{W_t^A/\kappa(n_t)\} + \exp\{W_t^0/\kappa(n_t)\}},$$

Here $\kappa(n_t) = \kappa_1$ if $n_t = 0$ and κ_C otherwise. Expected births equal $\pi_a \pi_{t,a}^F$.

Equilibrium

Given parameters, an initial population N_0 , and an initial household distribution G_0 , an **equilibrium** is a sequence

$$\{V_t, g_t, G_t, N_t, P_t, r_t, T_t, \varpi_t\}_{t \geq 0}$$

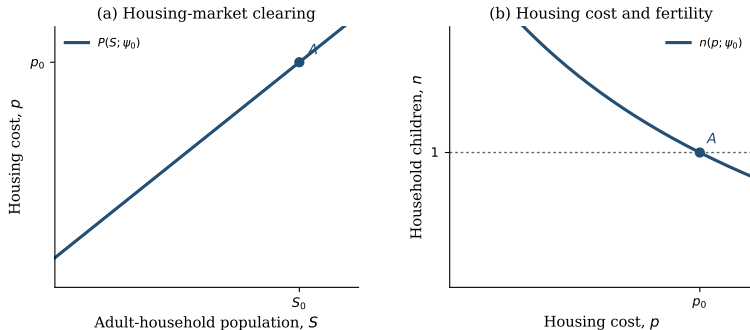
such that, for every t :

1. **Households.** Given $\{P_s, r_s, T_s, \varpi_s\}_{s \geq t}$, V_t solves the household problem with continuation V_{t+1} ; g_t attains it.
2. **Housing.** $r_t = (R_b + \delta_H + \tau_t^P)P_t - P_{t+1}$ and $\int q_t dG_t = H_0(P_t/\bar{P})^\eta$, with q_t the rooms occupied.
3. **Government.** Pensions balance payroll taxes; property-tax revenue is rebated equally:
$$\varpi_t \int_{a \geq a_R} dG_t = \tau^{SS} w \int_{a < a_R} e_a z_t dG_t, \quad T_t \int dG_t = \tau_t^P P_t \int h_{t+1} dG_t.$$
4. **Population.** Births $B_t = \int \pi_a \pi_{t,a}^F dG_t$ and $N_{t+1} = \mathcal{A}_t N_t + \mathcal{F}_t B_t + M_t$, where $\mathcal{A}_t$ ages surviving persons, $\mathcal{F}_t$ places surviving newborns at age zero, and M_t is net migration.
5. **Consistency.** G_{t+1} follows from G_t under g_t , earnings shocks, child maturation, and survival, with new households entering so that the number of households at each age is consistent with N_{t+1} .

Quantification

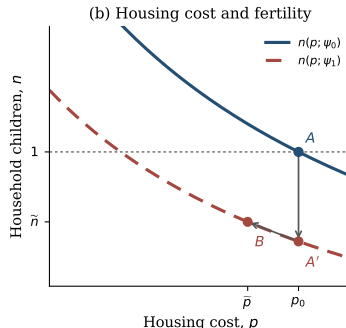
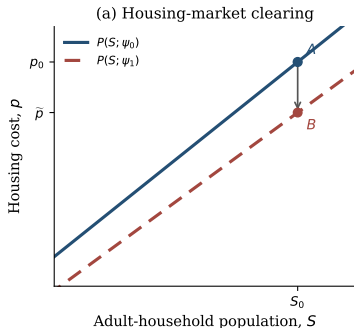
Initial Steady State

For intuition, summarize housing costs by p and fertility by $n(p; \psi)$.



At A , housing clears at (S_0, p_0) and fertility is at replacement (normalized to $n = 1$).

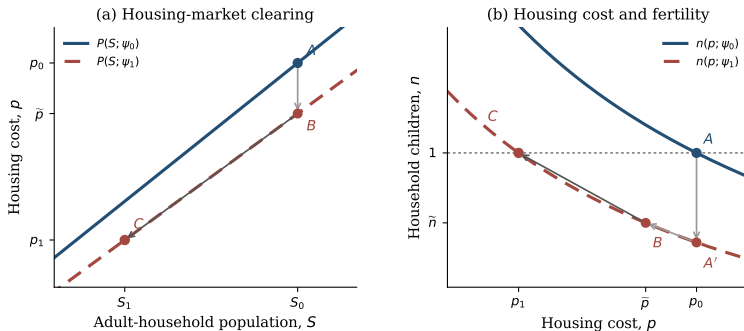
Impact



A fall from ψ_0 to ψ_1 lowers fertility at unchanged housing cost: $A \rightarrow A'$. Housing costs then adjust at inherited population S_0 : $A' \rightarrow B$.

These two movements decompose the response within the same period.

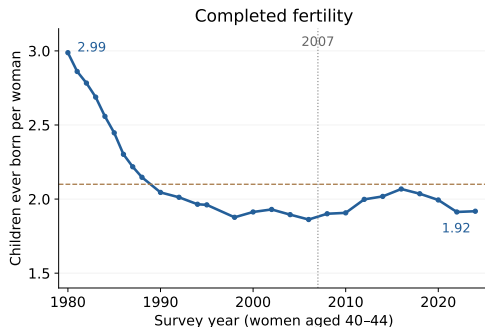
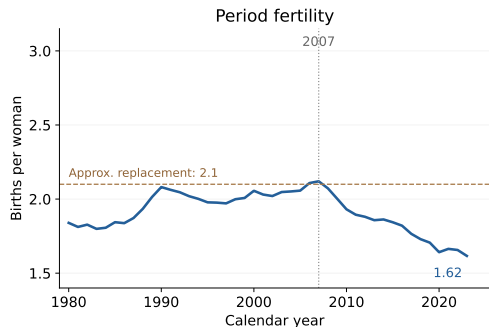
Demographic Adjustment



Smaller entering cohorts can reduce housing demand. Lower housing costs partly offset the preference decline; C marks a possible stationary outcome.

The arrow from B to C illustrates demographic adjustment. Changing age composition can produce overshooting; this is not a computed transition.

Fertility in the United States



We approximate the 2007 economy by an initial steady state.

Source: World Bank WDI; U.S. Census Bureau, CPS Historical Table 2.

Calibration and Historical Transition

1. Calibrate the 2007 steady state.

Match fertility, housing, and wealth moments to determine household parameters.

2. Estimate the sequence of fertility shocks.

Hold those parameters fixed and match birth rates through 2023 with unanticipated changes in fertility preferences. Carry households and population forward between shocks.

3. Project the economy forward.

After 2023, hold fertility preferences fixed and solve for prices, housing choices, and population. Compare housing policies from the same 2023 economy.

A model period is four calendar years; $\beta = \beta_{\text{annual}}^4$.

Utility Specification

For quantification, use the following functional form:

$$u_t(c_t, s_t; m_t) = \frac{\left[\frac{c_t^{\alpha_0} (s_t - \bar{h}(m_t))^{1-\alpha_0}}{e(m_t)} \right]^{1-\sigma}}{1-\sigma} + \psi_t m_t.$$

$$e(m_t) = \left(\frac{2 + 0.7m_t}{2} \right)^{0.7}, \quad \bar{h}(m_t) = h_P \mathbf{1}\{m_t > 0\}, \quad \alpha_0 = 0.733, \quad \sigma = 2.$$

Historical Fertility Discipline

Choose each preference level to match fertility over the following four calendar years.

Decision date	Birth years	Mean published TFR
2007	2008–2011	1.9749
2011	2012–2015	1.8610
2015	2016–2019	1.7554
2019	2020–2023	1.6458

The model sums age-specific births per adult household: an approximate analogue of female period TFR. The exposure mapping remains to be reconciled.

This is the estimation design. A complete historical fit requires converged conditional equilibria and stability to a longer continuation horizon.

Source: Published annual U.S. total fertility rates; equal-weight four-year averages.

Identification

Nine parameters are chosen jointly to match eleven moments, in three blocks.

Fertility

- ξ , first-birth cost $\rightarrow$ childlessness
- κ_1 , first-birth taste dispersion $\rightarrow$ mean age at first birth, first births after 30
- κ_C , later-birth taste dispersion $\rightarrow$ one-child families

Housing

- h_P , space requirement of parents $\rightarrow$ rooms added at first birth, rooms of 3+ versus 1–2 children
- χ , owner premium $\rightarrow$ ownership rate
- H_0 , supply scale $\rightarrow$ mean rooms

Saving and bequests

- β , discount factor $\rightarrow$ wealth over earnings
- θ_0 , bequest scale $\rightarrow$ bequest flow
- θ_1 , bequest wealth shift $\rightarrow$ old-age p90/median

The child preference ψ_0 is set so that completed fertility equals 2.1.

Calibration: Targets and Model

Moment	Target	Model
Completed fertility (normalization)	2.10	2.10
Childless, women 40–44 (%)	19.8	20.7
Exactly one child among mothers, 40–44 (%)	21.4	20.7
Mean age at first birth (years)	25.98	26.10
First births at age 30+ (%)	24.9	23.2
Mean occupied rooms (capped at 9)	5.56	6.29
Ownership, heads 30–55 (%)	64.8	52.4
First-birth room response, –1 to +3	0.720	1.011
Rooms: 3+ versus 1–2 resident children	0.347	0.200
Wealth / annual gross earnings	6.15	4.85
Annual bequests / wealth (%)	0.88	0.86
Wealth/income p90 / median, ages 76–84	3.52	4.44

Sources: CPS 2004/2006 (children ever born, women 40–44); natality records 2003–2006 (first-birth timing); ACS 2005/2006, 42 metros (rooms, ownership); PSID 2003/2005 (wealth); PSID event study (room response).

Internally Calibrated Parameters

Parameter	Economic interpretation	Value
β_{annual}	Annual discount factor	0.990
κ_1	First-birth taste dispersion	0.279
κ_C	Later-birth taste dispersion	0.347
χ	Owner housing-service multiplier	1.048
H_0	Housing supply at reference price	8.191
θ_0	Bequest-motive scale	0.087
θ_1	Bequest-utility wealth shift	0.057
ξ	First-birth utility cost	0.127
h_P	Parenthood space requirement	2.300
ψ_0	Initial preference for children	0.160

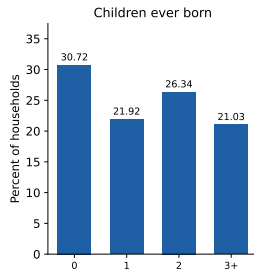
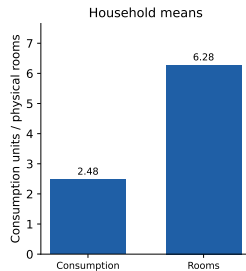
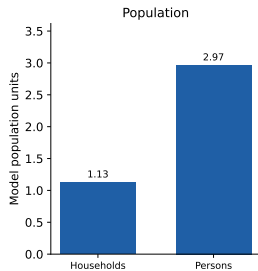
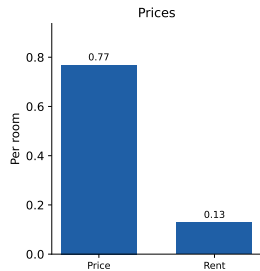
Externally Calibrated Parameters

Parameter	Value	Source
Risk aversion σ	2	
Equivalence scale $e(m_t)$	$((2 + 0.7m_t)/2)^{0.7}$	Scholz, Seshadri & Khitatrakun (2006)
Consumption share α_0	0.733	CEX childless cash renters, 2019–2023
Housing-supply elasticity η	0.63	
Annual real return i_{ann}	2.0%	Greaney et al. (2025)
Payroll tax rate τ^{SS}	17.9%	OECD U.S. average

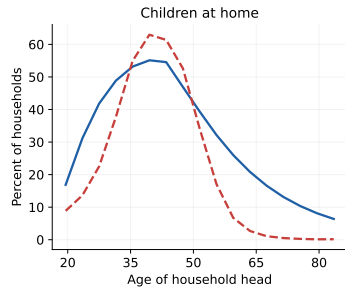
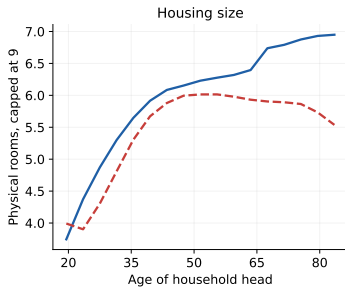
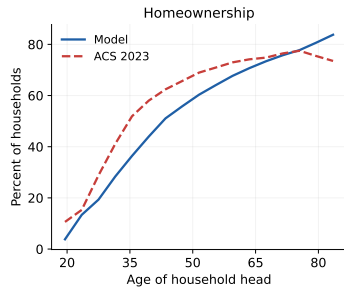
2023 Data and Model

Moment	Data	Model
Completed fertility, ages 40–44	1.92	1.70
Childless, ages 40–44 (%)	18.8	25.0
Exactly one child among mothers, 40–44 (%)	22.2	25.5
Mean age at first birth (years)	28.09	29.74
First births at age 30+ (%)	37.7	44.0
Mean occupied rooms (capped at 9)	5.58	6.07
Ownership, heads 30–55 (%)	58.7	46.9
First-birth room response, –1 to +3	0.720	1.060
Rooms: 3+ versus 1–2 resident children	0.355	0.266
Wealth / annual gross earnings	6.87	5.64
Annual bequests / wealth (%)	0.88	0.70
Wealth/income p90 / median, ages 76–84	3.45	3.50

The 2023 Equilibrium

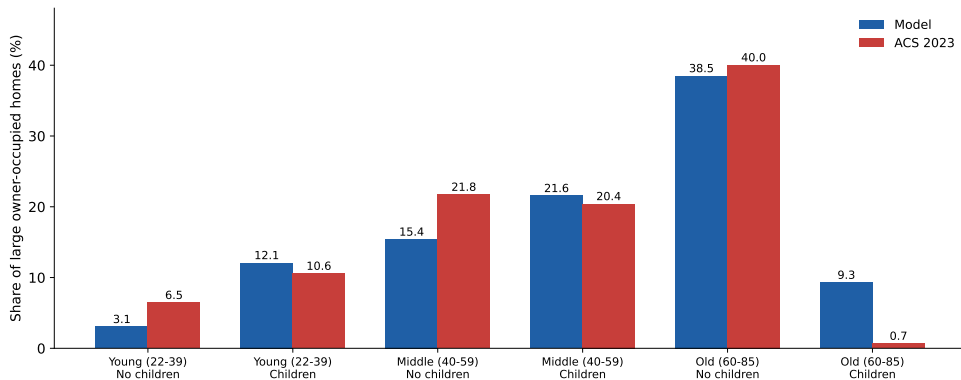


Lifecycle Fit in 2023



Children at home: model depends; ACS resident own children under 18.

Intergenerational Allocation: Model vs Data



Owner-occupied homes with 6+ rooms; ACS 2023. Children: model dependents; ACS resident own minors.

Stationary Housing Policy Comparisons

Compare three stationary equilibria under the same household parameters and fertility preference:

1. Annual property tax of 1%, with no rebate.
2. Annual property tax of 1%, with an equal rebate per household.
3. Annual property tax of 2%, with an equal rebate per household.

In the rebated cases, all property-tax revenue is returned to households. Payroll taxes separately finance pensions in each equilibrium.

Conditional on the diagnostic demographic and entry assumptions; these are stationary comparisons, not transition effects from 2023.

Stationary Policy Results

	1% tax No rebate	1% tax Equal rebate	2% tax Equal rebate
Period fertility	1.7298	1.8011	1.8080
Homeownership (%)	66.31	66.13	66.80
Rooms per household	6.734	6.626	6.298
House price per room	0.633	0.707	0.659
Rent per room, per period	0.105	0.117	0.136
Rebate per household, per period	0.000	0.187	0.332
Fertility change vs. no rebate (%)	0.00	+4.12	+4.52

With rebates in both cases, raising the tax from 1% to 2% increases fertility by 0.38%.

Conditional stationary comparisons under the same demographic and entry assumptions.

Conclusion

- Children raise demand for rooms; rental size and down payments govern access to that space.
- Tenure, saving, and fertility interact over the household lifecycle.
- Births change population and housing demand with long delays.
- Property taxes and rebates jointly change housing costs and household resources.

The stationary comparisons hold demographic and entry assumptions fixed.

Thank You

Appendix

1. Calibrating the Initial Economy

Goal: fit pre-2007 household moments. External inputs and all 12 targets and weights stay fixed.

1. **Try a parameter vector θ :** nine coefficients governing saving, fertility, housing, and bequests.
2. **Compute initial age-earnings shares** from survival, entry, and the earnings process. Set the pension to payroll revenue divided by retiree exposure; verify it against the solved distribution.
3. **Try a fertility preference ψ_0 and solve the housing equilibrium.** Guess the house price P , derive rent, solve household choices backward, and find the stationary distribution forward. Update P and repeat until housing demand equals supply.
4. **Update ψ_0 with a bracketed scalar search.** Repeat step 3 for each trial until model completed fertility equals 2.1 (within 0.0005).
5. **Score this parameter vector:** compute all 12 model moments and the weighted sum of squared model-target gaps.
6. **Update θ .** Perturb each coefficient to estimate how the moments respond; use these slopes for bounded, ridge-regularized joint steps. Redo steps 2–5 for each proposal, keep the lowest loss, and repeat within the search budget. Re-solve the selected winner twice.

Nested loops: parameter search $\longrightarrow$ fertility normalization $\longrightarrow$ housing-market clearing.

2. Fitting the Historical Transition

Hold the nine structural parameters fixed. Form 2007 age masses from data; within each age, keep the calibrated distribution of wealth, housing, and family states.

1. **Try ψ_{2007} , believed permanent.** Solve the terminal stationary economy at this preference level.
2. **Guess future price and pension paths.** Solve household choices backward from the terminal value; simulate household and person distributions and birth cohorts forward from the inherited 2007 state.
3. **Update the price and pension paths and repeat step 2** until housing clears and the payroll budget balances at every date. Verify the population accounting, recompute the path, and check a longer horizon.
4. **Change ψ_{2007} and repeat steps 1–3** until first-window fertility matches the 2008–2011 target of 1.974875 (error at most 0.005). Every trial starts from the same 2007 state.
5. **Advance only after acceptance.** Keep the first four-year outcome and birth queues. Repeat at 2011, 2015, and 2019 to target the following four years. Each new preference level is unanticipated.

After 2023, hold the final preference level fixed.

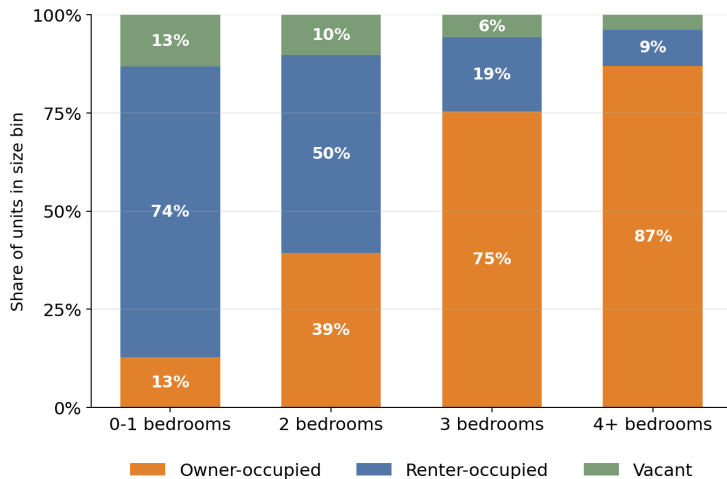
The household-rate analogue of female TFR is approximate; the complete history is not yet validated.

Parameter Restrictions

Parameter	Value	Lower	Upper	Position
β_{annual}	0.9900	0.94	0.99	Upper bound
κ_1	0.2794	0.02	50	Near lower bound
κ_C	0.3469	0.02	50	Near lower bound
χ	1.0480	0.1	5	Interior
H_0	8.1905	0.2	80	Interior
θ_0	0.0870	0	8	Interior
θ_1	0.0571	0.02	16	Near lower bound
ξ	0.1268	0	8	Interior
h_P	2.3000	0.1	2.3	Upper bound

Near-bound means within 1% of the allowed range; it does not establish a binding constraint. Fixed, normalized, and budget-derived objects are listed separately in the main table.

Big Units Are Owned



Source: American Housing Survey, 2023.

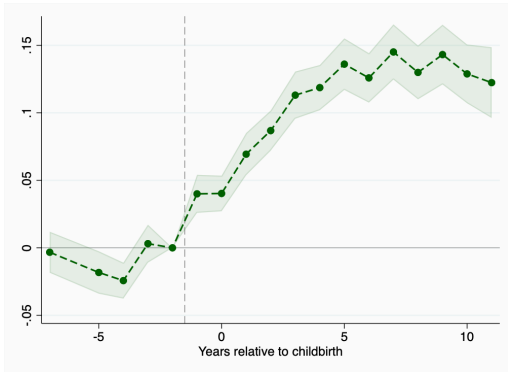
Data

- **AHS and ACS:** housing sizes, tenure, household composition, and age profiles.
- **PSID:** housing around childbirth, earnings, and wealth by age.
- **CPS and natality records:** completed fertility, childlessness, and birth timing.
- **Census population estimates:** age–sex population and demographic flows.

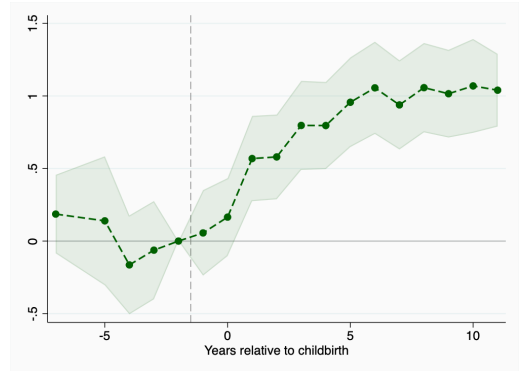
Cross-sectional housing patterns discipline access to space; longitudinal data describe housing adjustment around births.

Ownership and Housing Size Around First Birth

Ownership probability



Number of rooms

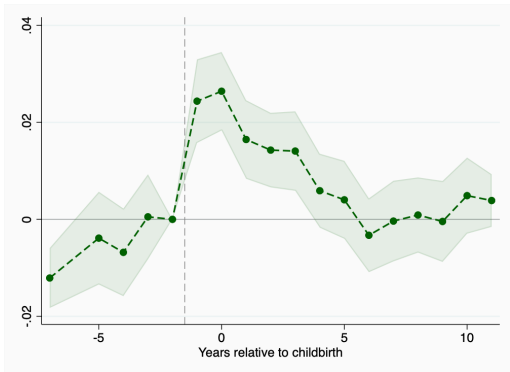


Original May event-study coefficients; measurement revisions under review.

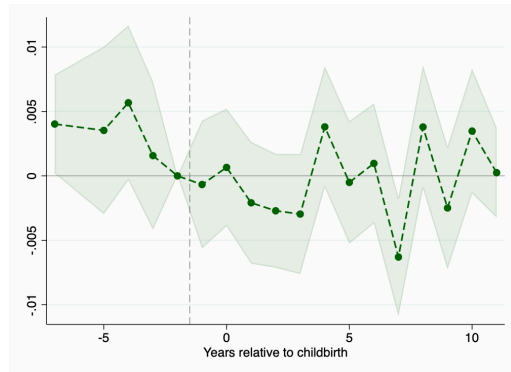
Source: PSID; original May event-study estimates.

Moving Around First Birth

Expansion or better housing



Neighborhood, schools, or proximity



Original May event-study coefficients; measurement revisions under review.

Source: PSID; original May event-study estimates.

Steady-State Equilibrium

A steady state is a constant allocation, population structure, and price:

$$(N^*, H^*, G^*, \pi^*, P^*, r^*, T^*, \varpi^*).$$

Households optimize, persons and household distributions reproduce themselves, housing clears, and the fiscal rule holds.

- With no migration, births must replace deaths in a constant population.
- With migration, births and net inflows together offset deaths.
- A constant continuation value alone does not make a finite path stationary.

The Person-Cohort Law

t indexes household periods; $j = 0, \dots, L$ indexes years within each period. $N_{t,j}^{g,\ell}$ counts persons of sex g and age ℓ in years.

For $j = 0, \dots, L - 1$, the annual person law is

$$N_{t,j+1} = A_{t,j+1}N_{t,j} + f_{t,j+1}B_{t,j+1} + M_{t,j+1}.$$

- $A_{t,j+1}$ ages existing cohorts using destination-age survival.
- $f_{t,j+1}B_{t,j+1}$ allocates newborn survivors by sex at age zero.
- $M_{t,j+1}$ is the age–sex net-migration flow after survival.
- Model-period births are assigned to calendar cohorts before the annual law is iterated.

The period boundary satisfies $N_{t+1,0} = N_{t,L}$; write $N_t = N_{t,0}$.

The Household-Head Law

Annual age–sex headship rates $\rho^{g,\ell}$ map persons into the target head stock:

$$H_{t,j+1}^{g,\ell} = \rho^{g,\ell} N_{t,j+1}^{g,\ell} = H_{t,j+1}^{\text{surv},g,\ell} + F_{t,j+1}^{g,\ell} - D_{t,j+1}^{g,\ell} + H_{t,j+1}^{M,g,\ell}.$$

Aggregating annual ages ℓ into household age cells a , the household operator supplies $\tilde{G}_{t+1,a}$. Its mass is rescaled to the person-law head target:

$$G_{t+1,a}(x_{t+1}) = H_{t+1,a} \frac{\tilde{G}_{t+1,a}(x_{t+1})}{\int \tilde{G}_{t+1,a}(\xi) d\xi}.$$

- F and D are the minimum one-way formation or dissolution flow needed to attain $\rho^{g,\ell}$.
- Wealth, tenure, income, and family composition are preserved within age.
- This is an accounting closure, not an estimated household-formation hazard.

Exact Annual Person Accounting

Let $s^{g,\ell}$ be survival and f^g the birth sex share. For newborns,

$$N_{t,j+1}^{g,0} = s_{t,j+1}^{g,0} f_{t,j+1}^g B_{t,j+1} + M_{t,j+1}^{g,0}.$$

For interior ages $\ell \geq 1$,

$$N_{t,j+1}^{g,\ell} = s_{t,j+1}^{g,\ell} N_{t,j}^{g,\ell-1} + M_{t,j+1}^{g,\ell}.$$

The terminal age is top coded, so survivors from the penultimate and terminal cells both enter the last cell. Every step satisfies

$$\sum N_{t,j+1} = \sum N_{t,j} + B_{t,j+1} - \text{newborn deaths} - \text{existing deaths} + \sum M_{t,j+1}.$$

The Within-Age Household Bridge

Let $\tilde{G}_{t+1,a}(x_{t+1})$ be the raw household transition within age a , and let $H_{t+1,a}$ be the head mass implied by persons. Then

$$G_{t+1,a}(x_{t+1}) = \begin{cases} H_{t+1,a} \frac{\tilde{G}_{t+1,a}(x_{t+1})}{\int \tilde{G}_{t+1,a}(\xi) d\xi}, & \int \tilde{G}_{t+1,a} > 0, \\ H_{t+1,a} \Gamma_a(x_{t+1}), & \text{for an empty entrant age cell,} \end{cases}$$

where Γ_a is the externally specified entrant template. The operation exposes added and removed mass separately and moves no mass across ages.

A richer model would estimate economic states for newly formed, dissolved, and migrant heads rather than preserve the raw conditional composition.

Housing-Market Clearing

Let $q_t(x_t)$ denote the physical housing occupied after tenure choice. Market clearing requires

$$Q_t^H = \int q_t(x_t) dG_t(x_t) = H_0 \left(\frac{P_t}{\bar{P}} \right)^\eta.$$

For renters, $q_t = h_t^R$; for owners, $q_t = h_{t+1}$. Choice probabilities are included when aggregating.

- The owner-service premium changes utility, not physical supply.
- Conditional renter policies enter demand only when renting is chosen.
- The same supply schedule applies to baseline and tax-reform paths.

Steady-State Solution

Fixed preferences and tax rules; no immigration.

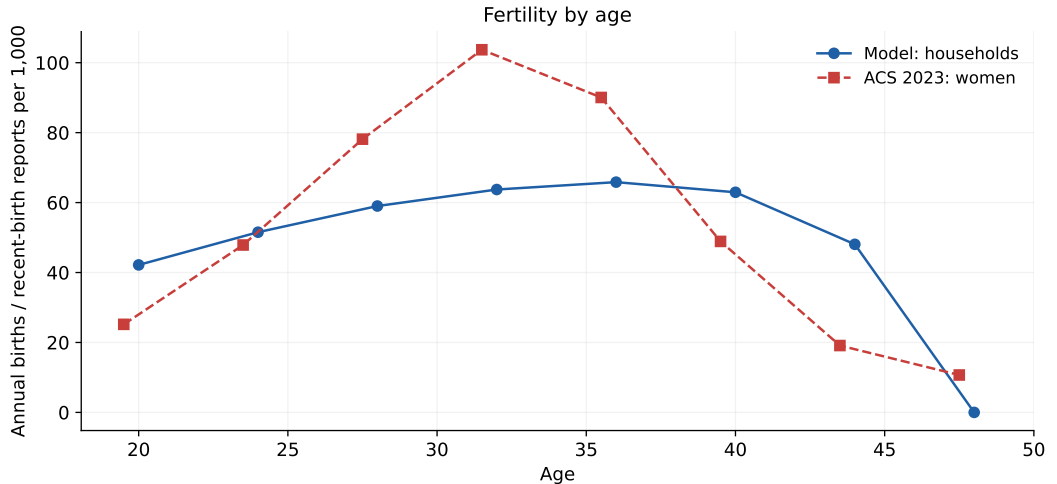
1. Guess the house price P , pension ϖ , and rebate T ; derive rent r from asset pricing.
2. Solve households backward over age: optimize saving and housing conditional on fertility and tenure, then compute choice probabilities.
3. Construct the stationary household composition. Set household mass equal to housing supply divided by mean housing per household.
4. Update P , ϖ , and T until household entry replaces exits, payroll taxes fund pensions, and property taxes fund equal rebates.
5. Apply the population law for one period and verify that the distribution and entry queue reproduce themselves; check markets and both budgets.

Transition Solution

Permanent preference change with a stationary terminal condition.

1. Fix inherited households and the entry queue; solve the new steady state for the terminal value.
2. Guess price, pension, and rebate paths (P_t, ϖ_t, T_t) ; derive rents from asset pricing.
3. Using these paths, solve households backward in calendar time.
4. Advance households and entry queues; aggregate housing demand and taxes.
5. Update all three paths jointly until housing clears and both fiscal budgets balance at every date.
6. Check terminal distributions and queues against the steady state; extend the horizon to test stability of early responses.

Fertility by Age: Measurement Comparison



Model: annualized 2023-2027 birth flow. ACS: women reporting a birth in the previous 12 months.

Prices and Quantities Along the Finite Path

